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Integrated lot-sizing and scheduling for parallel mixed-model automotive production lines with transportation resource constraints☆

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ABSTRACT

The automotive industry is undergoing rapid transformation, accompanied by intensifying market competition. As a result, component manufacturers face mounting pressure to improve production efficiency and control costs, particularly in high-mix, low-volume manufacturing environments. This paper investigates the integrated lot-sizing and scheduling problem in a mixed-flow parallel production system for automotive components, with a particular constraint related to the availability and management of transportation resources during production. The connection between lot-size scheduling and transportation resource allocation is crucial to the overall efficiency of the system. Therefore, a novel mixed integer programming model is presented for the multi-objective problem to minimize the makespan, total tardiness, total setup cost, and inventory levels. The min–max normalization approach is applied to normalize the objectives with different dimensions. A hybrid sparrow search algorithm (HSSA) is proposed with multi-strategy fusion, i.e., integrated mutation strategy, particle swarm velocity position update strategy, Gaussian Cauchy perturbation strategy, and multi-point crossover strategy. Two instances with varying scales have been designed to assess the performance of the proposed algorithm in the context of a practical automotive components production environment. The parameters of the HSSA for different sizes of problems are tuned by Taguchi method. Performance evaluation of the proposed HSSA is carried out by conducting extensive experiments compared to three well-established algorithms. The results demonstrate the effectiveness and superiority of the proposed HSSA in solving the simultaneous lot-sizing and scheduling problem with transportation resource constraints.

1. Introduction

With the development of the automotive industry and the increasing complexity of market demands, the manufacturing industry is facing unprecedented challenges. Mixed-model production (MMP), as a flexible and efficient production strategy, has been increasingly emphasized by academia and industry. MMP enables the simultaneous production of multiple product models on the same production line, meaning no changes to the main production equipment or process layout are required. Instead, quick adjustments and replacements of tools, molds, and other components can be made, allowing the same production line or processing unit to utilize the same technological processes and manufacturing methods to produce a variety of products. Mixed-model production involves adjusting the production sequence of various product varieties through scientific and effective maneuvers and producing them continuously in a rhythmic manner according to specific mixing ratios. Mixed-model production emphasizes product type and yield, enhancing resource utilization while effectively meeting diverse market demands. Mixed-flow production is significant in automotive component production because automotive manufacturing usually involves a variety of components, and its production process is complex and diverse [1]. The key points for optimizing the production are accurately determining the lot size of different orders, the production sequence, resource allocation, and equipment switching time.

This study focuses on the critical issue of scheduling with transportation resources, which play a pivotal role in the efficient management of material movement, caching, and transshipment operations under time-sensitive delivery constraints. The utilization and release of these resources have been shown to exert a substantial influence on the outcomes of scheduling. Conventional methods such as the Genetic Algorithms (GA) and the Particle Swarm Optimization (PSO) have been

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extensively employed, with numerous scholars delving into related concerns. For example, McMullen [2] proposed a heuristic approach through search to solve the mixed flow shop scheduling problem with setup time constraints. Pinedo [3] Classical scheduling theory covers mixed flow shop scheduling, resource allocation, and multi-objective optimization. Akpınar et al. [4] applied a hybrid genetic algorithm to optimize mixed-model assembly line configurations incorporating parallel workstations and partition constraints. However, given the timeliness constraints and the unique characteristics of transportation resources, designing a more efficient optimization method deserves further study.

1.1. Mixed-model production and lot-size scheduling

The mixed-model production model is widely used in automotive parts manufacturing to realize multi-variety and small lot production. The core challenge of mixed-model production is to balance the demand for different parts in the production process and avoid idle equipment or overloaded production line [5–8]. lot-size scheduling strategies are widely used to cope with diverse production demands [9]. The implementation of production scheduling optimization through sub-lot division and parallel processing sequencing, while adhering to production line capacity limitations and resource constraints, enables the reduction of equipment changeover duration and non-value-added downtime, thereby enhancing operational throughput efficiency in manufacturing systems [10,11]. lot-size scheduling reduces mold changeover time, enhances the production line's utilization, and optimizes the productivity. Scientific scheduling of production orders and parallel production can significantly reduce the switching time and equipment idle time [12]. He et al. [13] developed a novel algorithm to address the lot-size scheduling problem in distributed heterogeneous mixed-flow shop environments. The proposed algorithm effectively solves the multi-stage processing and lot re-entry problems of tasks, enhances the production process and boosts the overall efficiency of the production system. Huang et al. [14] proposed a two-stage, three-machine inter-vehicle problem based on the dynamic programming scheduling optimization method, significantly reducing the production cycle time and improving production efficiency. Yu et al. [15] proposed a multi-objective memetic algorithm for energy-efficient distributed assembly hybrid flowshop scheduling, optimizing tardiness and energy consumption.

While lot-size scheduling offers significant advantages in optimizing production efficiency, most existing studies rely on simplified assumptions, such as fixed resource allocation, and fail to analyze the complex relationships and priorities among orders comprehensively. Jabeur et al. [16] developed a reinforcement learning-based approach for integrated lot sizing and production scheduling under energy constraints, achieving better performance than conventional optimization methods in flexible flow line environments. Rohaninejad et al. [17] developed a satisfiability modulo theories-based approach for simultaneous lot-sizing and job shop scheduling under uncertainty, combining neural networks with K-means heuristic to optimize safety levels and achieving effective trade-offs between schedulability and optimality. Liu et al. [18] proposed a lot production model for a flexible job shop that accounts for setup time. They analyzed the coupling between lot-size scheduling and scheduling optimization and achieved more flexible production scheduling, improved production efficiency, and increased production capacity through an improved GA. Meanwhile, with the diversification of production demands, Kurniawan et al. [19] has been exploring refined scheduling strategies, such as task sequencing, priority assignment. However, these methods still face some challenges when facing the resource constraints of the production lines and frequent switching. For instance, Q. Luo et al. [20] indicated that in the production process of traditional Chinese medicine extraction facilities, lot size selection significantly affects production efficiency, manufacturing costs, and delivery times. Especially in the case of inconsistent order delivery, large lot production can reduce lead time, production efficiency, and manufacturing costs. However, it may lead to delayed delivery of some orders. Basir et al. [21] proposed a bi-level GA based on an extended MILP model, where the upper level sequences jobs and the lower level assigns them to lots. However, the algorithm faces high computational complexity for large-scale problems and its convergence and stability are sensitive to initial settings.

Extensive academic research has been dedicated to the lot-size scheduling problem, yielding a spectrum of solution frameworks to address its inherent complexities. Many studies have explored scheduling optimization strategies in different production modes and proposed classifications and solutions [22,23]. Avadiappan et al. [24] summarize the research results in lot-size scheduling, pointing out the strengths and weaknesses of the existing methods and future research directions. Amidst dynamic production demands, optimizing lot-size scheduling strategies has emerged as a critical research imperative.

1.2. Scheduling problems with transportation resource constraints

In the MMP of automotive components, transportation resources such as trolley are required during the whole production. Transportation resource allocation necessitates integrated functionalities encompassing intermediate buffering and transshipment within production. To guarantee the seamless execution of production orders, the scheduling of transportation resources needs to meet the constraints of time-sensitive delivery, and only after delivering the orders, the occupied transportation resources can be released [25]. Synchronously optimizing transportation resources in production scheduling becomes problematic in improving overall production efficiency. Especially under timeliness constraints, the scheduling of transportation resources should consider the priority of production tasks and ensure the maximum utilization of resources in the transportation process, thereby preventing conflicts and minimizing idle time between production and transportation processes.

Most existing research primarily concentrates on simply matching transportation resources with production tasks but ignores the cache timeliness constraints of transportation resources, the priority differences between production tasks [26], the matching relationships between transportation resources and product types, and the quantity capacity constraints of transportation resources. Geng et al. [27]developed a digital twin-based dynamic scheduling framework with multi-agent deep reinforcement learning (MAPPO-MC) to address discrete manufacturing workshop scheduling under transportation resource constraints and production disturbances. Dong et al. [28] proposed a heuristic-assisted deep Q-network (HA-DQN) algorithm for flexible job shop scheduling with AGV transportation resources, achieving 12.63% makespan reduction through cooperative optimization of job sequencing, machine selection, and AGV assignment. Zhang et al. [29] developed a multi-agent deep Q network framework for dynamic flexible job-shop scheduling with transportation time constraints, using reward-sharing mechanisms to address delayed rewards through collaborative machine and job agent optimization. Abderrahim et al. [30] developed a solution for the job shop scheduling problem in vehicle-based manufacturing facilities, focusing on optimizing the allocation of jobs to resources. Aminzadegan et al. [31] examined an integrated production and transportation scheduling problem designed to maximize order revenue and minimize the distribution through resource allocation and the associated processing time costs dependent on resources, weighted delay cost, and vehicle fixed cost. Fontes et al. [32] addressed a shop floor scheduling problem involving limited transport vehicles, proposing a PSO–simulated annealing hybrid algorithm to jointly optimize machine and vehicle scheduling, aiming to minimize completion and exit times. The problem involves two categories of resources: processing resources and transportation resources, where the processing resources are responsible for the manufacturing tasks of a

specific product and the transportation resources are accountable for delivering the parts to the processing area.

Moreover, Haj Mouldi et al. [33] studied the flexible job shop scheduling problem with multi-automatic mobile robot (AMR) transportation constraints. A GA-based solution is proposed to integrate scheduling rules to reduce the total completion time. Although the transportation resource scheduling model is proposed, it fails to fully consider the cache timeliness of transportation resources and the interactions between production tasks [34–37]. It can easily lead to inefficient resource utilization and delayed production tasks. The occupancy and release of transport resources are subject to strict time constraints, and resources can only be released and put back into use after meeting delivery requirements. This constraint makes the scheduling of transport resources more complex and variable, and balancing the utilization efficiency of transport resources and the sequence of production tasks becomes the key to improving the production system's overall efficiency.

1.3. Solution methodology

To address the scheduling challenges for mixed-model production systems, especially the complex scheduling problems with transportation resources and delivery requirements. Many scholars have done research in related fields. Wang et al. [38] proposed a feedback-based artificial bee colony algorithm for energy-efficient flexible flow shop scheduling. It improves global search through dual-swarm and dynamic strategies, effectively optimizing time, energy, and delay, with potential for broader application. Zeng et al. [39] proposed a method for solving lot job shop scheduling using mathematical modeling, an auction mechanism, and an improved graph model. Experiments show it outperforms other algorithms and CPLEX in both effectiveness and stability, especially on medium and large-scale problems. Li et al. [40] addressed the integrated optimization of worker assignment, lot splitting, and scheduling in a hybrid assembly-serial system. They proposed a MINLP model and a tailored VNS algorithm with layered coding. Li et al. [41] proposed a multidimensional co-evolutionary algorithm with Q-learning-based local search for hybrid flow shop scheduling, optimizing completion time and energy consumption under resource and robotic constraints. He et al. [42] proposed a hybrid ant colony optimization algorithm for shared scheduling of machines and AGVs in flexible manufacturing systems, combining heuristic vehicle allocation with production scheduling to avoid local optimums.production scheduling. Fathollahi-Fard et al. [43] developed an online mixed-integer programming model for sustainable distributed flow-shop scheduling, employing predictive-reactive strategies and decomposition heuristics to minimize makespan while constraining energy consumption in dynamic environments. Shahvari et al. [44] proposed a robust meta-heuristic algorithm combining local search and population structure to optimize the total weighted completion time and total weighted latency for a hybrid flow shop lot-size scheduling problem with sequential and machine-dependent family setup times. Besides meta-heuristic and swarm intelligence algorithms, machine learning-based approaches have also demonstrated promising potential in addressing various scheduling problems in industrial integrated manufacturing. Li et al. [45] proposed a graph reinforcement learning method with hierarchical aggregation neural network for dynamic scheduling in human-machine symbiosis manufacturing, incorporating subgraph cutting and attention mechanisms. Serrano-Ruiz et al. [46] developed a deep reinforcement learning approach using proximal policy optimization (PPO) for job shop smart manufacturing scheduling, modeling the environment as a Markov decision process with digital twin methodology. Additionally, recent research has increasingly focused on advanced optimization paradigms, particularly transfer learning, knowledge-assisted methodologies, and evolutionary multitasking frameworks, along with scheduling algorithms for cloud-fog-edge computing environments [47–51], which have demonstrated superior performance in dynamic resource allocation and multi-objective optimization scenarios. These state-of-the-art approaches represent significant advances in adaptive optimization techniques for complex scheduling and resource management problems.

Existing studies have developed effective domain-specific solutions, yet remain largely confined to production scheduling or transportation resource optimization paradigms. Although a variety of optimization algorithms have been proposed, these methods tend to ignore the timeliness of resource utilization and the handling of complex relationships among production lots, leading to difficulties in achieving optimal results when dealing with complex production tasks [52–54]. In contrast, the hybrid sparrow search algorithm with multi-strategy fusion proposed in this study integrates multiple factors such as the priority of production tasks, the caching time constraints of transportation resources, and the overall production efficiency. The combination of global optimization and local fine-tuning greatly improves the quality and computational efficiency of the scheduling scheme, and provides a new method for solving automotive component mixed-mode production [55] and lot-size scheduling problems.

Table 1 summarizes research gaps in mixed-model production scheduling from three aspects, positioning our work within industrial information integration literature. Existing studies typically address lot-sizing or transportation optimization in isolation, lacking systematic integration for automotive manufacturing contexts. While various methods (GA, PSO, ACO, deep reinforcement learning) have been proposed, they remain domain-specific with inadequate constraint integration. However, parallel mixed-model automotive production lines require comprehensive optimization of diverse components, transportation resources with timeliness and capacity quantity constraints, production varieties, delivery deadlines, and lot sizes.

The remaining content of this paper is structured as follows. Section 2 demonstrates the problem description. Section 3 comprises the proposed hybrid sparrow search algorithm with multi-strategy fusion. Section 4 indicates the computational experiments and results. Section 5 presents the engineering case study to validate the practical applicability of the proposed approach. Section 6 gives the conclusions and some future aspects of this research.

2. Problem formulation

This paper investigates an integrated lot-sizing and scheduling problem in a mixed-flow parallel production system with special transportation resource constraints, which originates from a real-world make to order (MTO) auto part company, as shown in Fig. 1. There are several parallel production lines in the workshop to produce semi-finished automobile parts, and during production, the trolleys are required to realize the parts transfer among processes and the temporary storage after the completion of the product. The parts are manufactured in lots, indicating that the different orders can be split or combined as long as certain conditions (i.e., job type, demand size, due date, required trolley type) are satisfied. In this manufacturing system, we first need to determine appropriate initial scheduling sub-lots for the orders and allocate them to parallel production lines evenly. When the sub-lot enters the production system, corresponding trolleys from the trolley resource pool are matched with the lot for full transportation and final temporary storage. If there are no matching trolley resources, the sub-lot will enter the sub-lots order pending queue. After the finished parts have been placed on the trolley for a certain period of time constraint, it is delivered downstream, and at the same time, the trolley resources are released and reused in the system. In the context of processing multiple product categories, frequent process alterations are imperative, leading to the mold changeover losses. These losses have two primary consequences including reducing equipment utilization and increasing order delivery risks by extending production cycles. Therefore, This

Table 1
Comprehensive literature review for the present study.

Research focus	Papers	Main contributions	Research gaps
Mixed-model production & lot-size scheduling	McMullen [2], Akpınar et al. [4], He et al. [13], Huang et al. [14], Yu et al. [15], Jabeur et al. [16], Rohaninejad et al. [17], Liu et al. [18], Basir et al. [21]	Lot-size optimization, setup time reduction, multi-objective optimization with energy efficiency, uncertainty handling	Simplified resource assumptions, fixed allocation strategies, simple order structure, no need preprocessing of order consolidation and batching
Scheduling with transportation resource constraints	Geng et al. [27], Dong et al. [28], Zhang et al. [29], Abderrahim et al. [30], Aminzadegan et al. [31], Fontes et al. [32], Haj Mouldi et al. [33]	Digital twin framework, multi-agent reinforcement learning, task scheduling with AGV, job-vehicle allocation	Strict association constraints between order items and transportation resource types, none consideration of cache time limitation, unified transportation resource attributes, fixed capacity constraints for transportation resource
Solution methodology	Wang et al. [38], Zeng et al. [39], Li et al. [40,41,45], He et al. [42], Fathollahi-Fard et al. [43], Shahvari et al. [44], Serrano-Ruiz et al. [46]	Artificial bee colony algorithm, auction mechanisms, multi-dimensional co-evolutionary approach, hybrid ant colony optimization, deep reinforcement learning, graph neural networks	Domain-specific solutions, difficult to solve the highly coupled problem with multiple features and constraints, such as mixed flow system, parallel production lines, order preprocessing, lot-sizing, multi-functional transportation resources, cache time limitation of transportation resources

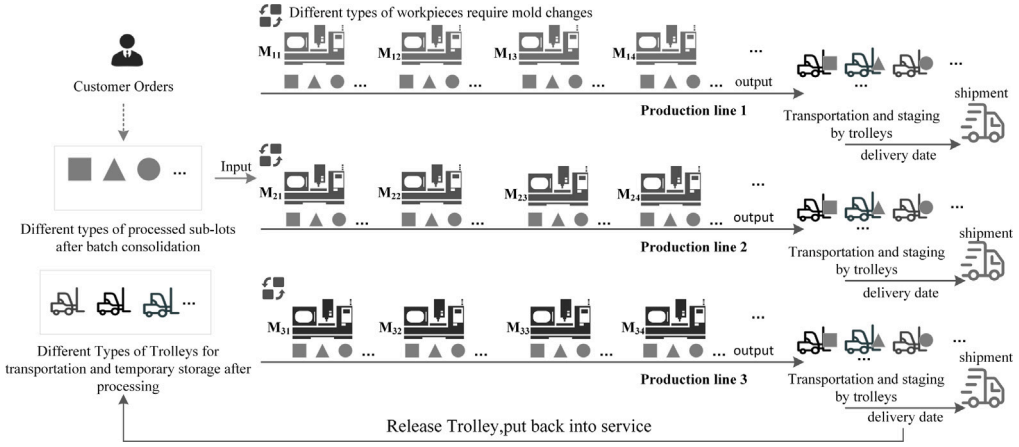


Fig. 1. Configuration of parallel mixed-model automotive production lines with transportation resource constraints.

paper focuses on the integrated lot-sizing and scheduling for unrelated parallel production lines with dual-function transportation resource constraints, to achieve delivery requirements and efficient production. The notations and decision variables employed in this study are defined in Table 2.

2.1. Basic assumption

Since current research is motivated by the practical production scenarios, it exhibits unique characteristics tailored to the specific business needs of the enterprise. For example, during the production, the trolley is needed for transferring the parts among processes and temporarily storing the sub-lots. The number of trolleys constraint directly affects the transfer speed of parts. Moreover, insufficient trolleys may lead to production line blockage, affecting the overall production progress. A mold change operation is required when the production line produce different types of products continuously, meanwhile a setup time occurs. Frequent changeovers will directly lead to the waste of time and lower production efficiency. To ensure more efficient production, lower inventory costs and fewer switching times, It is very important to make optimal scheduling for the production system, including four coupled subproblems:lot-size settings for different orders, sub-lots distribution among the parallel lines, the trolley resources coordination and sub-lots scheduling on each line. In formulating a mathematical model for such a practical scheduling problem, several assumptions need to be clarified as below.

i. All trolleys are idle at the initial moment and can be used at any time.

ii. All mold change times are fixed and uninterruptible.

iii. All workpieces are assumed to arrive at time$t = 0$ without considering the waiting time of the production line initially.

iv. Each processing equipment processes at most one workpiece at a time and the workpiece processing cannot be interrupted.

v. All production lines are ready for operation at t=0.

vi. Transportation, loading and unloading times associated with the trolley can be ignored in this system.

Table 2
Notations of the problem.

Symbols	Description of model parameter definitions
D	Types of trolley, $d = \{1, 2, \ldots, D\}$
L	Production lines, $l = \{1, 2, \ldots, L\}$
J	Workpieces, $j = \{1, 2, \ldots, J\}$
W	Types of workpiece, $w = \{1, 2, \ldots, W\}$
O	Orders, $o = \{1, 2, \ldots, O\}$
H	The sub-lots obtained through pre-processing of the orders, $h = \{1, 2, \ldots, H\}$
K_l	The sub-lots assigned to production line l , $k_l = \{1, 2, \ldots, K_l\}$
h_o^w	The workpiece type of sub-lot h from the order o is w
DD_h	Delivery time for the sub-lot h
J_{ho}	The workpieces in sub-lot h from order o, $j_{ho} = \{1, 2, \ldots, J_{ho}\}$
TF_l	Unavailable time period for production lines l
$B_{j_{ho}}$	The processing start time for workpieces j_{ho}, $j_{ho} = \{1, 2, \ldots, J_{ho}\}$
$E_{j_{ho}}$	The processing end time for workpieces j_{ho}, $j_{ho} = \{1, 2, \ldots, J_{ho}\}$
S_l	Categories of workpieces compatible with processing capabilities of the production line l
τ_{wl}	Time required for a workpiece of type w on production line l
F_{tw}	Quantity of finished goods in stock for workpieces of type w at time t, $w = \{1, 2, \ldots, W\}$
R_{tw}	Number of new finished workpieces of type w at time t, $w = \{1, 2, \ldots, W\}$
U_d	Number of workpieces of type w that can be stored in a trolley of type d
V_d	A set of workpiece types which can be loaded in the trolley of type d
Z_d	Number of the trolleys of type d
c_j	Work-in-process inventory cost per unit of workpiece j per unit of time
Y_{hkl}	0-1 Variable, if the sub-lot h is processed on production line l then Y_{hkl} is 1, otherwise 0
$Minls$	Minimum lot size for the processing sub-lots
$Makespan$	Maximum completion time of all the orders
TT	The total tardiness of the orders
TNS	Total number of switches
IC	Total cost of the inventory
ρ_1	Weight of $Makespan$ in the objective function
ρ_2	Weight of TT in the objective function
ρ_3	Weight of TNS in the objective function
ρ_4	Weight of IC in the objective function
f	Objective function value-Total weighted objective

vii. The processing time of each workpiece on each production line is a given value.

2.2. Mathematical model formulation

Makespan is a crucial objective that indicates the efficiency of a production system, particularly in multi-variety and small-lot manufacturing environments. In this research, makespan is the longest completion time among the parallel production lines. The makespan is given by Eq. (1). Total tardiness is another important objective that measures the overall delay in a production system, particularly in environments where timely delivery is crucial. The total tardiness objective is expressed in Eq. (2). The frequency of mold switchovers is minimized to prevent excessive capacity wastage using Eq. (3). Inventory cost is another crucial objective to minimize using Eq. (4). It reflects the expenses associated with holding inventory in the system. A weighted sum method is adopted here to transform multiple objectives into a single objective, that is, the total weighted objective (*TWO*), and it is presented in Eq. (5).

$$minf_1 = Makespan = \left\{\max\left(C_{\max}^{l}\right)\right\}, \forall l = (1, 2, \ldots, \mathrm{L}) \tag{1}$$

$$minf_2 = TT = \sum_{\mathrm{h}=1}^{H}\sum_{j=1}^{J_{ho}}\left[\max\left\{\left(E_{J_{ho}} - DD_h\right), 0\right\}\right] \tag{2}$$

$$minf_3 = TNS = \sum_{l=1}^{L}\sum_{k=1}^{K-1}\left|\frac{\sum_{h=1}^{H}\left(Y_{hkl}\cdot H_o^w\right) - \sum_{h'=1}^{H}\left(Y_{h',k+1,l}\cdot H_o^{\prime w}\right)}{S_l}\right|, \quad \sum_{h'=1}^{H} Y_{h',k+1,l} = 1 \tag{3}$$

$$minf_4 = IC = \sum_{j=1}^{J_{ho}}\sum_{h}^{H}\left(DD_h - B_{j_{ho}}\right)\cdot c_j \tag{4}$$

$$minf = \sum_{i=1}^{4}\rho_i \cdot f_i \tag{5}$$

To ensure that the model can accurately reflect the characteristics of the mixed-flow production system in the actual scenario, we must thoroughly consider all kinds of constraints when constructing the scheduling model. These constraints cover the material transfer resources, the number of trolleys, the mold change, the processing capacity of equipment and the order of lots in the production flow.

$$\sum_{k=1}^{K}\sum_{l=1}^{L} Y_{hkl} = 1, \forall h = \{1, 2 \ldots \ldots, H\} \tag{6}$$

$$\sum_{h=1}^{H}\sum_{k=1}^{K}\sum_{l=1}^{L} Y_{hkl} = H \tag{7}$$

$$Y_{hkl} = 0, h = \{1, 2\ldots\ldots, h\}, w_h \notin S_l \tag{8}$$

$$\sum_{h=1}^{H} Y_{hlk} \geq \sum_{h=1}^{H} Y_{hl,k+1}, l = \{1, 2, \ldots ...L\}, k = \{1, 2, \ldots ..., K-1\} \tag{9}$$

$$B_{j_{ho}} \geq 0, j_{ho} = \{1, 2, \ldots ..., J_{HO}\} \tag{10}$$

$$E_{j_{ho}} = B_{j_{ho}} + \sum_{l=1}^{L}\sum_{k=1}^{k}\left(Y_{hd}\cdot\tau_{wl}\right) \tag{11}$$

$$B_{j_{ho}} + \sum_{k=1}^{K}\left(Y_{hkl}\cdot\tau_{wl}\right) \notin TF_l, j_{ho} = \left\{1, 2 \ldots \ldots J_{HO}\right\}$$
$$B_{j+1,ho} \geq E_{j_{j_b}}, j_{ho} = \left\{1, 2, \ldots \ldots J_{HO} - 1\right\}, h = \{1, 2, \ldots \ldots .H\}, \quad o = \{1, 2 \ldots \ldots, O\} \tag{12}$$

$$\sum_{h=1}^{H}\left(Y_{hkl}\cdot E_{hkl}\right) \leq \sum_{h=1}^{H}\left(Y_{h'kl}\cdot E_{hkl}\right), h \neq h', k = \{1, 2 \ldots \ldots ..K-1\}$$
$$\& \sum_{h'=1}^{H}\left(Y_{h'kl}\cdot E_{hkl}\right) > 0 \tag{13}$$

$$R_{tw} = \sum_{h=1}^{H} \sum_{j=1}^{J_{ho}} \left(\left[1 - \left| \frac{E_{j_{h_o}} - t}{E_{j_{h_0}} + t} \right| \right] \cdot \left[1 - \left| \frac{w_h - w}{w_h + w} \right| \right] \right), \quad w = \{1, 2, \ldots \ldots, W\} \tag{14}$$

$$F_{tw} + R_{tw} \leq Z_d \cdot U_d, w \in V_d \tag{15}$$

$$\sum_{j=1}^{J_{ho}} J_{ho} \geq Minls, \forall j \tag{16}$$

Eq. (6) is an order exclusivity constraint, i.e., the processing sub-lot of orders must be assigned to one and only one production line. Eq. (7) indicates all the sub-lots must be scheduled on parallel production lines. Eq. (8) denotes that the sub-lot must be assigned according to the production lines capable. Eq. (9) represents that the sub-lots on each production line must be scheduled consecutively, with no empty positions between sub-lots. Eq. (10) ensures that the start time of the sub-lot h from order o should be greater than or equal to 0. Eq. (11) gives the computing method about the finish time of the workpiece J_{ho} with consideration of the unavailable time of the production line. Eq. (12) ensures that the start time of the next workpiece within the same sub-lot is greater than or equal to the completion time of the preceding workpiece. Eq. (13) implies that for two adjacent sub-lots on the same production line, the start time of the first workpiece in the next sub-lot is greater than or equal to the completion time of the last workpiece in the previous sub-lot. Eq. (14) is used to calculate the number of workpieces of type w coming off the line at time t. Eq. (15) limits that the number of products produced at any moment cannot exceed the limit of the total number of pieces that can be loaded in the corresponding trolley to which it belongs (The workpiece type and the trolley type have mutually specified constraint relationships). Eq. (16) shows that any sub-lot size in the parallel lines system must satisfy the constraint of minimum production lot size.

3. Proposed hybrid sparrow search algorithm

3.1. Framework of HSSA

In this paper, we present an improved sparrow search algorithm for the discussed problem. The standard sparrow search algorithm (SSA) is a swarm intelligence optimization algorithm based on nature's behavior, initially introduced by Xue et al. [56]. The algorithm simulates the collaboration and individual exploration abilities demonstrated by sparrows during the foraging process, aiming to seek the optimal global solution through efficient collective intelligence. Compared with the traditional swarm intelligence algorithms, SSA significantly improves search efficiency and convergence speed by introducing the cooperation and competition mechanism among sparrows. The core mechanisms of SSA include foraging behaviors, following behaviors, and an early warning mechanism, which correspond to different population updating strategies in the algorithm. It can balance global exploration and local exploitation during the search process. The forager is responsible for discovering potential high-quality solutions in the search space with the position update using Eq. (17).

$$x_{i,j}(t+1) = x_{i,j}(t) \cdot \exp\left(-\frac{i}{\alpha \cdot T}\right) + r \cdot N(0,1) \tag{17}$$

Where t is the current iteration count, T denotes the maximum iteration count., α is a tuning parameter, r is a random number, and $N(0,1)$ is a normal distribution with mean 0 and variance 1. The follower optimizes the current solution by following the forager, whose position update method is described by Eq. (18).

$$x_{i,j}(t+1) = x_{\text{best},j}(t) + \beta \cdot \left| x_{i,j}(t) - x_{\text{best},j}(t) \right| \tag{18}$$

Where $x_{best,j}(t)$ is the position of the current global optimal individual in the jth dimension, and β is a random factor located in the interval [0,1]. When the sparrow senses danger in the external environment

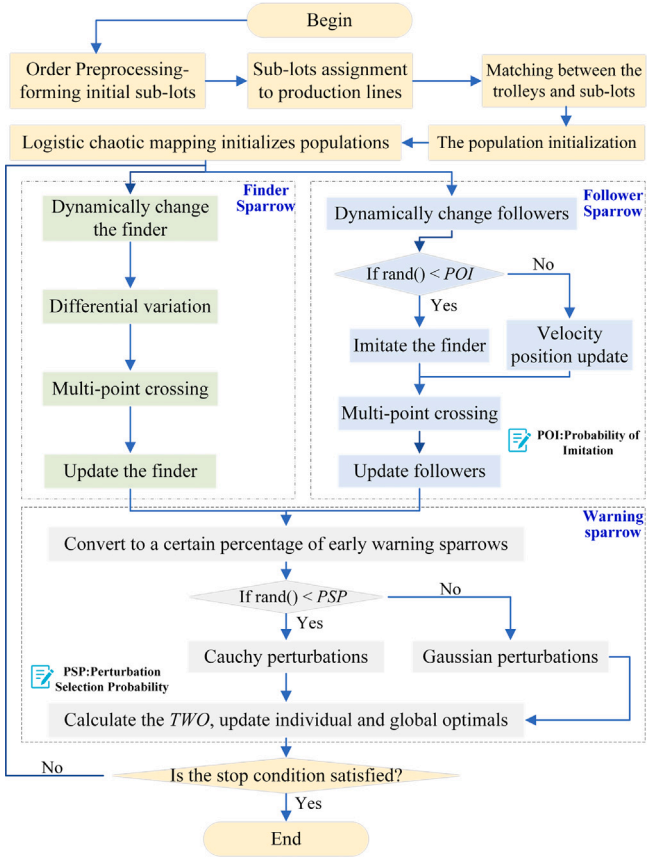


Fig. 2. Flowchart of hybrid sparrow search algorithm (HSSA) with multi-strategy fusion.

(e.g., a drastic change in the gradient of the objective function), it triggers an early warning behavior that forces the population to move to a more optimal solution. At this time, the individual positions are updated using Eq. (19).

$$x_{i,j}(t+1) = x_{i,j}(t) + q \cdot \text{rand} \cdot \left(x_{i,j}(t) - x_{\text{worst},j}(t) \right) \tag{19}$$

Where q is the step factor, $rand$ is a random number in the interval [0,1], and $x_{worst,j}(t)$ denotes the current position of the worst individual.

The overall framework of the algorithm designed in this study integrates heuristic rules with a multi-strategy sparrow search algorithm for automotive component mixed-flow parallel production line scheduling with dual-function transportation resource constraints, and the flowchart of the HSSA is illustrated in Fig. 2. The algorithm first performs rule-based order preprocessing to form sub-lots and considers preferred and alternative line configurations to assign them to production lines. Subsequently, trolley matching allocates appropriate transportation resources based on each sub-lot's type, delivery time, and ensuring sufficient trolley capacity for sub-lot transportation. To enhance population diversity and search efficiency, logistic chaotic mapping is employed to initialize the population, replacing conventional random initialization methods. In the HSSA framework, finder sparrows are responsible for discovering promising solutions in the scheduling landscape, follower sparrows explore alternative solution spaces to diversify the search process, and warning sparrows serve as "guardians against convergence stagnation", thereby preventing entrapment in local optima.

The designed multi-strategy framework incorporates four key enhancement mechanisms for automotive component mixed-flow parallel production line scheduling with dual-function transportation resource

Algorithm 1 Order preprocessing rule

1. **Input:** orders, due_dates, workpiece_types
2. **Output:** sub_lots, lot_labels
3. **Begin**
4. Sort the orders by the EDD rule
5. Combine the orders with same workpiece type and same delivery time
6. Set lead time threshold
7. Combine the orders with same workpiece type and within the lead time range
8. Set the minimum sub-lot quantity threshold
9. **for** each merged lot **do**
10. **if** lot size > max_lot_size **then**
11. Split it into smaller lots by *Minls*
12. **else if** lot size < min_lot_size **then**
13. Combine with the nearest lot and make sure the combined lot-size
14. does not exceed *Minls*
15. **end if**
16. **end for**
17. Label the final sub-lots
18. **return** sub_lots, lot_labels
19. **End**

constraints. Specifically, finder sparrows employ differential evolution strategies to update solutions while simultaneously utilizing multi-point crossover operations to enhance exploration of feasible scheduling sequences. Follower sparrows adopt velocity-position update strategies from particle swarm optimization for solution refinement, complemented by multi-point crossover strategies to strengthen search performance across different production scenarios. Subsequently, specific individuals in the population are selected according to a predetermined proportion, and warning sparrows probabilistically select either Gaussian perturbation or Cauchy perturbation to maintain solution diversity and prevent convergence into local optimization. This multi-strategy integration ensures comprehensive exploration of the scheduling solution space while maintaining convergence efficiency toward optimal production sequences.

3.2. Order preprocessing

3.2.1. Preprocessing and encoding

The order batching is depend on the number of workpieces in the order, the processing time of the workpieces, the delivery time of the order, and the constraint of the trolley. The following aspects can generally determine whether the orders need batching or splitting. Workpieces within the same sub-lot do not require switching time, while workpieces in different sub-lots with different type take longer to switch. A threshold is set for the upper limit of the workpieces number of sub-lot, and if it is less than the threshold, the workpieces can be combined into one lot. Furthermore, a threshold value is set for the difference in delivery time of workpieces within the same sub-lot, if it is less than a specific value, it can be batched into one lot.

Now suppose the sub-lots after preprocessing, is shown in Table 3. In this table, lot number represents the identification code of the processing sub-lots after lot merging, type of workpiece indicates the type of processing sub-lots, lot size refers to the required product quantity within the processing sub-lots obtained after lot merging operations based on thresholds set according to the enterprise's production capacity and operational experience, and due date represents the time calculated from the moment when the original orders are received to the delivery date.

Based on the above lot-sizing constraints, we propose an order preprocessing strategy to form the sub-lots for all the released orders. The pseudo-code for the order preprocessing rule is presented in Algorithm 1.

Step 1: All initial orders are sorted according to the earliest due date (EDD) rule, and then the orders with the same workpiece type and the same delivery date are combined in a big lot.

Table 3
The sub-lots information after order preprocessing.

Sub-lot number	Type of workpiece	Lot_*size*	Lot_*minls*	Due date
1	E	360	360	990
2	E	240	360	990
3	E	320	360	990
4	A	180	360	1470
5	B	360	360	1470
6	D	360	360	1470
7	D	200	360	1470
8	E	360	360	1470
9	E	300	360	1470
10	C	360	360	1950
11	C	320	360	1950
12	C	360	360	1950

Step 2: Set a lead time threshold value and combine the orders for the same type of workpieces within the lead time range in the order of sorting in step 1.

Step 3: Set the minimum quantities for the processing sub-lots. The big lots exceeding this minimum value will be split into smaller sub-lots, while those below the minimum will be combined with the nearest orders with same-type workpiece through forward shifting.

Step 4: The final sub-lots are labeled in order to be scheduled in the following processes of the algorithms.

All the sub-lots are sorted by the earliest delivery priority rule during individual initialization. Sub-lots with the same type and within the same due date range can be grouped. The sub-lots with different types may be arranged randomly. An example of this initialized population structure is illustrated in Fig. 3.

3.2.2. Decoding

Individuals in the group are entered into a simulation program based on heuristic rules for decoding. These sub-lots are allocated to production lines and trolleys according to the following principles.

- For the sub-lots with same workpiece type, priority is given to assigning it to the same production line.
- For the sub-lots with different workpiece type, select the corresponding production line for processing using the earliest completed production line principle. Suppose multiple production lines are available to process the sub-lot. In that case, the production line with the shortest completion time is selected. Only if there is just one production line can process the sub-lot, the sub-lot is directly distributed on that line.

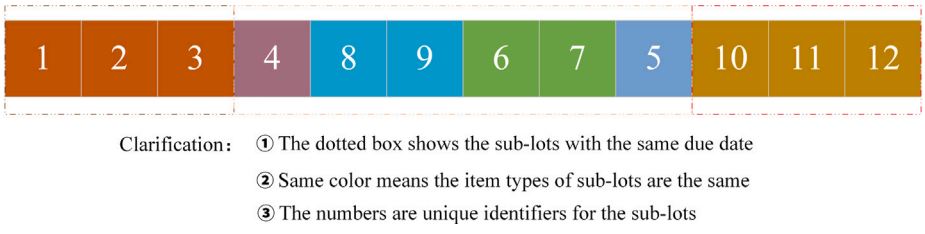


Fig. 3. Encoding representation.

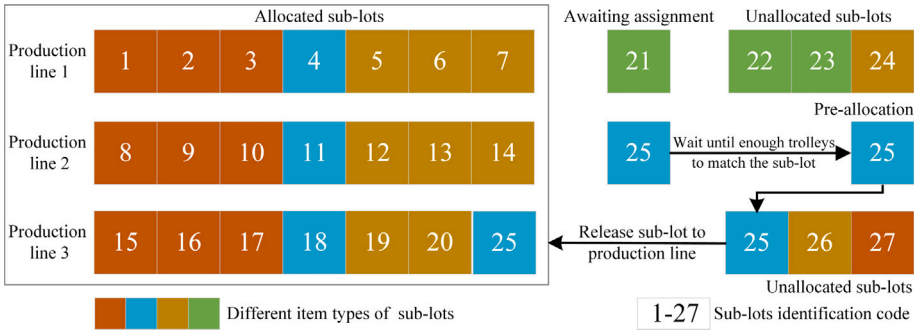


Fig. 4. Decoding diagram for the sub-lots releasing to the production line.

- Before allocating the sub-lots, the trolley constraints must be considered. If the number of trolleys corresponding to the sub-lot is insufficient, the lot must wait until the sufficient specified trolleys are released back to the resource pool, which is shown in Fig. 4.

The initialization of the chromosome processing sub-lots follows an ascending order of delivery, with lots of the same variety grouped. The simulation program allocates processing sub-lots and ensures that sub-lots of the same variety are assigned to the same line. Specific allocation steps are as follows and the decoding process is presented in Algorithm 2.

Algorithm 2 Sub-lot releasing rule

```
Input: sub-lots, production_lines, max_trolleys
Output: production_schedule
Begin
Sort sub-lots by due date, group same varieties
Initialize pre_allocated_set = ∅, used_trolleys = 0
while unallocated sub-lots exist do
  if trolley constraint violated then
    Move sub-lot to pre_allocated_set
  else
    Assign to same-type line or earliest-completion line
    Update used_trolleys
  end if
  if delivery date reached then
    Release trolleys, pre_allocated_set
  end if
end while
return production_schedule
End
```

Step 1: Before assigning the unallocated sub-lot to the available production line, make sure whether the total required number of trolleys exceed the available number of trolleys. If not, allocate the trolleys to the sub-lot and make deductions from available pool of this type of trolley. If the trolley limit has been exceeded, do not make the assignment at this time. Instead, insert the sub-lot at the end of the pre-allocated order set and go to check the next sub-lot.

Step 2: Determine whether the sub-lot to be allocated is the same type as the last sub-lot in the processing queue of the available production line. If it is the same, allocate it to the production line firstly, otherwise prioritize the allocation to the production line with a short completion time and transfer to step 3.

Step 3: When a sub-lot reaches its standing time and due date in the finished product buffer, the order is delivered and the corresponding trolleys are released to the production lines. update the number of available trolleys and check the pre-allocated order set. If it is not empty, all the orders in the pre-allocated set are sequentially inserted in front of the first order in the unallocated set, and the process is transferred to step 1.

3.3. Proposed HSSA with multi-strategy fusion

3.3.1. Logistic chaos mapping initialization

In the traditional SSA, the population initialization is usually generated by a random method, which may lead to an uneven population distribution, affecting the algorithm's global search ability. This study uses logistic chaos mapping (LCM) to improve the initialization phase of SSA and the population variety and quality of the initial solution. LCM is a simple and classical model of chaotic systems with initial value sensitivity and traversal, capable of generating uniformly distributed and complex chaotic sequences. By introducing LCM into the population initialization process, the search space can be covered more efficiently, enhancing the global search capability of the algorithm and accelerating the convergence speed. The formula for the logistic chaos map is shown in Eq. (20).

$$z_{k+1} = r \cdot z_k \cdot \left(1 - z_k\right), z_k \in (0,1), r \in (3.57, 4] \tag{20}$$

Where z_k denotes the kth value of the chaotic sequence, r is a control parameter, when $r = 4$ the mapping is fully chaotic, and z_0 is the initial value $z_0 \in (0,1)$. Let us assume that the target search space for SSA population initialization is $\Omega \subseteq R^d$ and its upper and lower bounds are $[L, U]$. Conventional random initialization of the population distribution obeys a uniform distribution $U[L, U]$ and with the introduction of LCM, after initialization, the position of each individual in the population is determined by Eq. (21).

$$x_{i,j} = L_j + z_{(i-1)\cdot d+j} \cdot (U_j - L_j), \quad i = 1, 2, \ldots, n; j = 1, 2, \ldots, d \tag{21}$$

In this linear transformation, $z_{(i-1)\cdot d+j}$ denotes the chaotic value generated by the logistic mapping sequence for the jth dimension of the ith individual, which is then mapped to the search space $[L_j, U_j]$. Here, i denotes all individuals (candidate solutions) in the population and j represents each individual's jth dimension of the problem.

3.3.2. Differential variation and particle swarm velocity-position updating strategies

Population intelligence optimization by simulating the behaviors of discoverers and followers in foraging. Despite its strong global search capability, it may face the problem of degraded search efficiency and is prone to getting trapped in local optima in complex high-dimensional optimization problems. This study introduces differential variation strategy and particle swarm speed-position update strategy in the discoverer and follower stages to enhance the algorithm's versatility and global exploitation ability. This helps to improve the computational efficiency and solution quality of the algorithm. The discoverer is responsible for the global search to explore potential quality solution regions. To further enhance the exploratory ability of the discoverer, this study introduces the differential variation strategy into this phase. Differential variation is an operator derived from the differential evolution (DE) algorithm. Its core idea is to generate new candidate solutions by adding weighted difference vectors to the current solution, as shown in Eq. (22).

$$x_i' = x_i + F \cdot \left(x_{r1} - x_{r2}\right) \tag{22}$$

Where, x_i' is the updated solution for the$i$th discoverer, x_i is the current solution, x_{r1} and x_{r2} are two distinct individuals randomly selected from the current population. Two different individuals are chosen randomly in the population; F is the difference coefficient of variation, which controls the magnitude of the difference vector and usually takes values in the range $F \in [0,2]$. The follower is mainly responsible for local exploitation in the potential high-quality area provided by the discoverer. This study introduces the speed-position update mechanism of the PSO algorithm in the follower stage to enhance the local search capability. Its computing model is given by Eq. (23).

$$v_i^{t+1} = \omega \cdot v_i^t + c_1 \cdot r_1 \cdot \left(p_i - x_i^t\right) + c_2 \cdot r_2 \cdot \left(g - x_i^t\right) \tag{23}$$

The v_i and v_i^{t+1} are the velocity of the$i$th follower at generation t and $t+1$; ω is the inertia weight, which controls the global–local balance of the search; c_1 and c_2 are the learning factors, which weigh the effects of the individual's best position p_i and global best position g on the update, respectively; $r_1 r_2 \sim U(0,1)$ is a uniformly distributed random numbers for increasing the randomness of the population; x_i^t is the current position of the ith follower. The position is updated according Eq. (24).

$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{24}$$

The improved SSA population update consists of two main phases, i.e., discoverer phase global exploration and follower phase (localized development) calculated using Eq. (25) & Eq. (26), respectively.

$$x_i^{t+1} + x_i^t + F \cdot \left(x_{r1}^t - x_{r2}^t\right) \tag{25}$$

$$v_i^{t+1} = \omega \cdot v_i^t + c_1 \cdot r_1 \cdot \left(p_i - x_i^t\right) + c_2 \cdot r_2 \cdot \left(g - x_i^t\right), x_i^{t+1} = x_i^t + v_i^{t+1} \tag{26}$$

3.3.3. Fusion of Gaussian and Cauchy perturbations for the scout(Warner sparrows)

In the SSA approach, the main task of an early warning agent is to avoid potentially hazardous regions through a randomization strategy to enhance the survival of the population. However, the traditional randomization strategy usually lacks specificity, which may lead to declining population diversity or getting trapped in the local optimum. For this reason, this paper introduces Gaussian perturbation and Cauchy perturbation in the scout stage to enhance the randomization and robustness of the population. Gaussian perturbation is a strategy based on the normal distribution, where the core idea is to simulate random movements in nature by applying small perturbations to the positions of individuals, thereby enhancing the local exploration capabilities of the population. Its mathematical expression is shown in Eq. (27).

$$x_i' = x_i + \epsilon_{\mathrm{G}} \tag{27}$$

Where x_i is the current individual position, $\epsilon_G \sim N\left(0, \sigma^2\right)$, is a random perturbation obeying a normal distribution, and σ is the standard deviation, determining the magnitude of the perturbation. Besides, the Cauchy perturbation leverages the long-tailed property of the Cauchy distribution to give individuals a greater random jumping ability, thereby expanding the search range and preventing convergence to a local optimum. Its mathematical expression is given in Eq. (28). Where $\epsilon_C \sim N\left(0, \sigma^2\right)$ is a random perturbation obeying a Cauchy distribution, μ is a position parameter, and γ is a scale parameter. Improved perturbation strategies for early warning agents are assessed using Eq. (29).

$$x_i' = x_i + \epsilon_C \tag{28}$$

$$x_i' = \begin{cases} x_i + \epsilon_G, & \text{if } r < p \\ x_i + \epsilon_C, & \text{if } r \geq p \end{cases} \tag{29}$$

3.3.4. Continuous to discrete mapping methods

Since the solutions generated by the strategies used in the HSSA with multi-strategy fusion are continuous type, however, the current shop floor scheduling problem is discrete combinatorial optimization characteristics. Therefore, this paper designs an efficient continuous-to-discrete mapping method to transform the solutions generated by the continuous optimization algorithm into discrete solutions conforming to the scheduling constraints. The proposed technique effectively ensures the uniqueness and complete coverage of the discrete solution through group ordering and missing correction strategy. First, the continuous solution vector p is divided into G approximately equal-length subgroups of length $\lceil L/G \rceil$ each according to its total length L. The subgroups are then divided into two subgroups of length $\lceil L/G \rceil$. The consecutive values within each subgroup are sorted according to their size, and the sorting result is mapped to the corresponding discrete integer index. The formula is computed by Eq. (30).

$$P_g = \mathrm{sort}\left(P_{\mathrm{indices}}\right), d_g = \mathrm{indices}\left(P_g\right) \tag{30}$$

Where P_g is the continuous value and d_g is the discrete index within the current group. The group ordering ensures that continuous values are reasonably mapped to discrete values in a localized range, initially completing the continuous to discrete conversion. In the second step, de-duplication and missing correction, the discrete solution needs to be further corrected to satisfy the uniqueness constraints of the shop floor scheduling problem, as the grouping strategy may lead to duplicate or missing indexes between different subgroups. The correction steps include uniqueness extraction, which extracts the unique set of integers U from the initial discrete solution, and missing value complementation, which fills the unassigned target set of integers M at the duplicate indexes. Eq. (31) gives the mathematical expression.

$$d = \mathrm{correct}\left(d_{\mathrm{group}}\right) \tag{31}$$

Where *correct*() is the correction function that performs the de-duplication and complementation operations. In summary, the steps of the method are to input a vector of continuous solutions p, sort and discrete map p into groups to generate an initial discrete solution, perform de-weighting and missing corrections to ensure the legitimacy of the solution, and output a final discrete solution d. The group sorting effectively maintains the localization of the continuous solutions, which helps generate high-quality discrete solutions. The correction process ensures that all the integer values appear uniquely in the solution, which satisfies the stringent constraints of the shop floor scheduling problem.

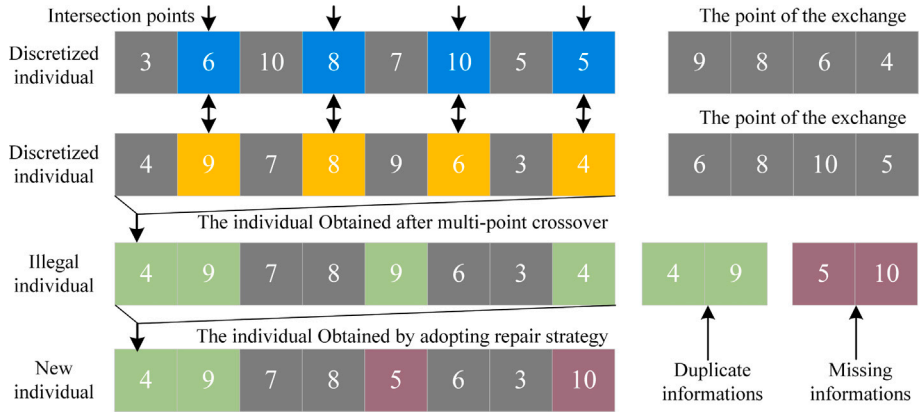


Fig. 5. Multi-point crossover.

3.3.5. Multi-point crossover

A multi-point crossover strategy is an improved solution generation method, as shown in Fig. 5. By randomly selecting multiple locations for exchanging values, multi-point crossover strategy can realize the diversity of solutions and the inheritance of high-quality solutions, which enhances the local search ability and avoids getting trapped in the local optimum. Assuming that the length of the current solution is n, the multi-point crossover strategy consists of the following steps, randomly determining the number of points n_{swaps} to be swapped based on the problem size. The mathematical description is Eq. (32). Where $\alpha \in [0.02, 0.08]$ is the ratio of intersection points. Subsequently, the set of indexes of the intersections $\mathcal{J}$ is randomly selected using the Eq. (33). For each selected index $i \in J$, the corresponding position of the new solution is inherited from the parent solution, as shown in Eq. (34). For unselected positions $j \in J$, the original solution is retained. The mathematical description is Eq. (35).

$$n_{\text{swaps}} = randint\,(2, \lceil \alpha \cdot n \rceil) \tag{32}$$

$$\mathcal{J} = \text{randperm}\left(n, n_{\text{swaps}}\right) \tag{33}$$

$$x'_i = x_i^{\text{parent}} \tag{34}$$

$$x'_j = x_j^{\text{original}} \tag{35}$$

4. Computational experiment

There are three parallel production lines to manufacture a key component with 12 different types for the automobiles in the real-world auto parts workshop. 7 different types of trolleys are recycled in the production system. Two categories of test cases containing large-scale and small-scale are generated according the actual production scenarios, in order to evaluate the performance of the proposed HSSA with a multi-strategy fusion sufficiently. The types of trolleys, along with their quantity and capacity, are shown in Table 4. Due to the fact that each production line operates in the flow line mode, each production line can be simplified into one machine according to its tact time during scheduling. Therefore, the studied parallel production line scheduling problem can be regarded as a parallel machine scheduling model. The correspondence between part types and production lines is indicated in Table 5.

Table 4
Trolley types, quantities, and capacities.

No.	Trolley type	Capacity	Quantity
1	IN	6	118
2	XR	10	250
3	CI	10	240
4	UR	6	450
5	AV	6	100
6	EN	10	100
7	EL	8	104

Table 5
Production line selection constraint for different type of parts.

Product type	Trolley type	Production lines preferred	Production lines alternative
item-1	IN	3	1, 2
item-2	IN	3	1, 2
item-3	XR	1, 3	2
item-4	CI	2, 3	1
item-5	CI	2, 3	1
item-6	UR	1, 3	2
item-7	AV	1, 3	2
item-8	EN	2, 3	1
item-9	XR	1, 3	2
item-10	EL	3	1, 2
item-11	UR	1, 3	2
item-12	UR	1, 3	2

4.1. Parameter tuning for proposed HSSA

Algorithm performance is highly sensitive to parameter configuration. To ensure robust performance and validate the effectiveness of the proposed HSSA, comprehensive parameter sensitivity analysis and optimization are conducted. This section presents systematic experiments to determine optimal parameter settings and demonstrates the algorithm's superiority through comparative analysis. Based on the enterprise's actual production capacity and operational experience, the experimental validation is conducted on two dataset scales: small-scale problems representing typical daily production scenarios with approximately 3000 orders, and large-scale problems representing peak production periods with approximately 5000 orders. Both scales encompass orders with 12 types to ensure comprehensive evaluation across varying problem complexities that reflect real-world manufacturing conditions. According to the original order preprocessing rules mentioned in Section 3.1, after order preprocessing, the small-scale problems have approximately 60 sub-lots waiting to be fed into the production lines, while the large-scale problems have approximately 100 sub-lots waiting to be fed into the production lines. Due to significant differences in multi objective dimensions, min–max normalization is used to normalize the objectives into a single objective. The original and normalized objective calculation function are shown in Eqs. (36) and (37).

$$\min f = \rho_1 \cdot f_1 + \rho_2 \cdot f_2 + \rho_3 \cdot f_3 + \rho_4 \cdot f_4 \tag{36}$$

Table 6
The parameters value at each level of 6 factors.

Level	Perturbation probability	Gaussian scale	Cauchy scale	Discoverer variation rate	Multi-point crossover proportion	Scout proportion
1	0.4	0.5	5	0.15	0.05	0.05
2	0.45	0.6	6	0.2	0.08	0.08
3	0.5	0.7	7	0.25	0.11	0.10
4	0.55	0.8	8	0.3	0.14	0.13
5	0.6	0.9	9	0.35	0.17	0.15

Table 7
Orthogonal array (OA) for Taguchi design of experiments for HSSA.

Experiment	Perturbation probability	Gaussian scale	Cauchy scale	Discoverer variation rate	Multi-point crossover proportion	Scout proportion
1	0.4	0.5	5	0.15	0.05	0.05
2	0.4	0.6	6	0.3	0.17	0.08
3	0.4	0.7	7	0.2	0.14	0.1
4	0.4	0.8	8	0.35	0.11	0.13
5	0.4	0.9	9	0.25	0.08	0.15
6	0.45	0.5	5	0.3	0.11	0.15
7	0.45	0.6	6	0.2	0.08	0.05
8	0.45	0.7	7	0.35	0.05	0.08
9	0.45	0.8	8	0.25	0.17	0.1
10	0.45	0.9	9	0.15	0.14	0.13
11	0.5	0.5	5	0.2	0.17	0.13
12	0.5	0.6	6	0.35	0.14	0.15
13	0.5	0.7	7	0.25	0.11	0.05
14	0.5	0.8	8	0.15	0.08	0.08
15	0.5	0.9	9	0.3	0.05	0.1
16	0.55	0.5	5	0.35	0.08	0.1
17	0.55	0.6	6	0.25	0.05	0.13
18	0.55	0.7	7	0.15	0.17	0.15
19	0.55	0.8	8	0.3	0.14	0.05
20	0.55	0.9	9	0.2	0.11	0.08
21	0.6	0.5	5	0.25	0.14	0.08
22	0.6	0.6	6	0.15	0.11	0.1
23	0.6	0.7	7	0.3	0.08	0.13
24	0.6	0.8	8	0.2	0.05	0.15
25	0.6	0.9	9	0.35	0.17	0.05

$$\min f = \rho_1 \cdot \frac{f_1 - f_1^{\min}}{f_1^{\max} - f_1^{\min}} + \rho_2 \cdot \frac{f_2 - f_2^{\min}}{f_2^{\max} - f_2^{\min}} + \rho_3 \cdot \frac{f_3 - f_3^{\min}}{f_3^{\max} - f_3^{\min}} + \rho_4 \cdot \frac{f_4 - f_4^{\min}}{f_4^{\max} - f_4^{\min}} \tag{37}$$

Parameter sensitivity analysis reveals that algorithm performance is significantly influenced by key control parameters, particularly in complex optimization scenarios. Traditional trial-and-error approaches for parameter tuning are computationally expensive and often fail to identify optimal parameter combinations due to parameter interactions. To address this challenge, this study employs the Taguchi experimental design method to systematically determine the optimal parameter configuration for the HSSA.

Through preliminary sensitivity analysis, six critical parameters are identified as having substantial impact on algorithm convergence behavior and solution quality: the perturbation probability of the scout, Gaussian perturbation step size, Cauchy perturbation step size, Discoverer Variation rate, Multi-point crossover proportion, and the scout proportion. The Taguchi method offers significant advantages over full factorial designs by reducing experimental runs while maintaining statistical rigor, making it particularly suitable for multi-parameter optimization problems.

A 5-level, 6-factor orthogonal array is designed based on Taguchi method to accommodate both large-scale and small-scale production requirements. Table 6 presents the orthogonal design matrix with all parameter combinations. Table 7 shows the comprehensive parameter matrix, yielding 25 distinct parameter combinations for experimental validation, with six replications conducted for each combination to ensure statistical reliability. Three additional key parameters were required to be set based on the practical conditions of manufacturing systems, including population size, maximum iteration count, and inertia weight. The population size is 200 and the maximum iteration limit is 1000. Furthermore, multi-objective weighting coefficients for small-scale and large-scale instances are independently calibrated according to production line operational requirements.

Six experiments were performed on each group of parameters, and Table 8 shows the results of numerical experiments. The statistical averages and standard deviation for each parameter group were independently derived, with the corresponding outcomes comprehensively indicated in Table 9. The fluctuations and distinctions observed through the 25 experimental groups were graphically illustrated in Fig. 6 and Fig. 7 to enhance the interpretability of intricate datasets. This analysis further explores the relationship between parameter adjustments and algorithm efficacy. The six parameters are analyzed using range analysis method based on the results of the numerical experiments in Table 10 and Table 11 for both scale scenarios. These two tables show the summation and average value of *TWO* at the 5 levels for the 6 factors. Based on these datas, the range value of the 6 factors can be obtained, where $L_i(\mathbf{i} = 1, \cdots, 5)$ denotes sum of *TWO* for each factor at level i. $\overline{L_x}(x = 1, \cdots, 5)$ represents the average value corresponding to level x. Consequently, the mean experimental index for each factor at each level indicates its influence on the experimental outcomes. The results derived from range analysis in Table 10 and Table 11 indicate that the proportion of early-warning sparrows exerts the most significant impact on algorithm performance under small-scale instance. While for large-scale instance, the multi-point crossover percentage has the most significant impact on algorithm performance. According to the sensitivity information from Figs. 6 and 7, the optimal values of the six factors for different size of instances can be selected and indicated in Table 12.

Table 8
The numerical results for the two kind of instances.

Small scale	Total weighted objective						Large scale	Total weighted objective					
	Test1	Test2	Test3	Test4	Test5	Test6		Test1	Test2	Test3	Test4	Test5	Test6
1	0.1053	0.1036	0.1148	0.1209	0.1206	0.1201	1	0.0893	0.0907	0.0924	0.1142	0.1057	0.1012
2	0.1325	0.1299	0.1084	0.1130	0.1121	0.1296	2	0.1223	0.0951	0.1200	0.1177	0.1227	0.1232
3	0.1391	0.1121	0.1390	0.1344	0.1106	0.1115	3	0.1266	0.1135	0.1147	0.1156	0.1196	0.1111
4	0.0900	0.1205	0.1301	0.1109	0.1106	0.1341	4	0.1212	0.1005	0.1243	0.1023	0.1226	0.1138
5	0.1393	0.1233	0.1119	0.1302	0.1163	0.1213	5	0.1098	0.0973	0.1190	0.1251	0.0961	0.1174
6	0.1392	0.1299	0.1298	0.1201	0.1128	0.1392	6	0.0948	0.1090	0.0989	0.1066	0.1233	0.1129
7	0.1214	0.1113	0.1296	0.1052	0.1391	0.1296	7	0.1128	0.1138	0.1045	0.1036	0.1082	0.1147
8	0.1110	0.1150	0.1209	0.1263	0.1201	0.1328	8	0.1002	0.1191	0.1287	0.1190	0.1181	0.1129
9	0.1142	0.1217	0.1047	0.1083	0.1205	0.1214	9	0.1201	0.1170	0.1181	0.1161	0.1022	0.1192
10	0.1202	0.1393	0.1227	0.1210	0.1300	0.1202	10	0.1217	0.1286	0.1204	0.1229	0.1239	0.1126
11	0.1109	0.1129	0.1216	0.1417	0.1122	0.1007	11	0.1183	0.1223	0.1190	0.1037	0.1189	0.1126
12	0.1303	0.1184	0.1202	0.1202	0.1298	0.1239	12	0.1230	0.1236	0.1199	0.1092	0.1203	0.1180
13	0.1070	0.1201	0.1304	0.1202	0.1199	0.1127	13	0.1254	0.1197	0.1112	0.1068	0.1191	0.1148
14	0.1345	0.1295	0.1026	0.1105	0.1201	0.1129	14	0.1139	0.1245	0.1032	0.1170	0.1191	0.0871
15	0.1300	0.1239	0.1310	0.1027	0.1215	0.1125	15	0.1144	0.1240	0.1211	0.1225	0.1157	0.1013
16	0.1391	0.1298	0.1390	0.1387	0.0973	0.1314	16	0.1129	0.1005	0.0851	0.1133	0.1026	0.1029
17	0.1346	0.1129	0.1218	0.1178	0.1259	0.1070	17	0.1150	0.1155	0.1113	0.1184	0.1151	0.1101
18	0.1107	0.1119	0.1424	0.1265	0.1414	0.1304	18	0.1197	0.1185	0.1117	0.1155	0.1210	0.1211
19	0.1119	0.1119	0.1241	0.1303	0.1298	0.1308	19	0.1129	0.1162	0.1060	0.1231	0.1174	0.1152
20	0.1084	0.1178	0.1278	0.1205	0.1299	0.1138	20	0.1264	0.0972	0.1163	0.1210	0.1201	0.0997
21	0.1107	0.1110	0.1109	0.1334	0.0919	0.1118	21	0.1120	0.1238	0.1220	0.1084	0.1107	0.1133
22	0.1203	0.1248	0.1410	0.1295	0.1298	0.1298	22	0.1217	0.1228	0.1145	0.1111	0.1210	0.1102
23	0.1296	0.1248	0.1153	0.1410	0.1161	0.1184	23	0.1166	0.1107	0.1129	0.1031	0.1216	0.1154
24	0.1298	0.1045	0.1215	0.1099	0.1108	0.1344	24	0.1153	0.1081	0.1108	0.1139	0.1000	0.1251
25	0.1308	0.1114	0.1167	0.1353	0.1262	0.1276	25	0.1081	0.1223	0.1178	0.1170	0.1122	0.1122

Table 9
Experimental results of Means and standard deviations for the two kind of instances.

Small scale instance	Total weighted objective		Large scale instance	Total weighted objective	
	Average value	Standard deviation		Average value	Standard deviation
1	0.1112	0.0077	1	0.0989	0.0090
2	0.1209	0.0099	2	0.1168	0.0099
3	0.1244	0.0132	3	0.1169	0.0050
4	0.1160	0.0146	4	0.1141	0.0096
5	0.1237	0.0090	5	0.1108	0.0109
6	0.1285	0.0096	6	0.1076	0.0093
7	0.1227	0.0116	7	0.1096	0.0044
8	0.1210	0.0071	8	0.1163	0.0086
9	0.1151	0.0067	9	0.1155	0.0061
10	0.1255	0.0070	10	0.1217	0.0048
11	0.1167	0.0127	11	0.1158	0.0061
12	0.1238	0.0047	12	0.1190	0.0048
13	0.1184	0.0072	13	0.1162	0.0061
14	0.1183	0.0110	14	0.1096	0.0119
15	0.1203	0.0100	15	0.1165	0.0076
16	0.1292	0.0148	16	0.1029	0.0094
17	0.1200	0.0089	17	0.1142	0.0028
18	0.1272	0.0126	18	0.1179	0.0034
19	0.1231	0.0082	19	0.1151	0.0051
20	0.1197	0.0075	20	0.1134	0.0110
21	0.1116	0.0120	21	0.1150	0.0058
22	0.1292	0.0063	22	0.1169	0.0052
23	0.1242	0.0090	23	0.1134	0.0057
24	0.1185	0.0109	24	0.1122	0.0076
25	0.1247	0.0082	25	0.1149	0.0046

4.2. Computational results

In this subsection, the proposed integrated lot-sizing and scheduling problems were tested on generated small-scale instances (data01-data06) after simplified linearization of the model, using Cplex solver 12.10.0.0. Table 13 shows the computational results. Within the 2-hour computation time limit, the Cplex12.10.0.0 solver can optimally solve instances with sub-lots up to 15 (data01-data02 with optimality gap of 0%). However, as the problem scale increases, Cplex's performance deteriorates significantly. For data03 with 20 sub-lots, Cplex achieves a 0.45% optimality gap; for data04 with 25 sub-lots, Cplex only obtains a poor solution with a 34.7% optimality gap after 7200 s; for data05 with 30 sub-lots, the optimality gap reaches 23.3% within the time limit; and for data06 with 35 sub-lots, Cplex completely fails to find any feasible solution. In contrast, the HSSA algorithm successfully obtains high-quality solutions for all test instances within 63 s, demonstrating remarkable efficiency. Notably, HSSA achieves significantly better total weighted objective compared to Cplex in larger instances: 0.1984 vs 0.1993 for data03, 0.1900 vs 0.2913 for data04, and 0.2420 vs 0.3157 for data05. Therefore, the Cplex solver is deemed inadequate for solving industrial-scale problems with larger numbers

Table 10
Range analysis of 6-factor 5-level Taguchi experiment for small-scale instance.

	Perturbation probability	Gaussian scale	Cauchy scale	Discoverer variation rate	Multi-point crossover proportion	Scout proportion
L1	0.5963	0.5973	0.6038	0.6115	0.5910	0.6001
L2	0.6129	0.6166	0.5978	0.6020	0.6182	0.5916
L3	0.5974	0.6153	0.6125	0.5888	0.6118	0.6183
L4	0.6193	0.5911	0.6137	0.6170	0.6085	0.6025
L5	0.6082	0.6139	0.6160	0.6147	0.6046	0.6217
avgL1	0.1193	0.1195	0.1208	0.1223	0.1182	0.1200
avgL2	0.1226	0.1233	0.1196	0.1204	0.1236	0.1183
avgL3	0.1195	0.1231	0.1225	0.1178	0.1224	0.1237
avgL4	0.1239	0.1182	0.1227	0.1234	0.1217	0.1205
avgL5	0.1216	0.1228	0.1232	0.1229	0.1209	0.1243
Range value	0.0046	0.0051	0.0036	0.0056	0.0054	0.0060

Table 11
Range analysis of 6-factor 5-level Taguchi experiment for large-scale instance.

	Perturbation probability	Gaussian scale	Cauchy scale	Discoverer variation rate	Multi-point crossover proportion	Scout proportion
L1	0.5575	0.5402	0.5782	0.5650	0.5582	0.5548
L2	0.5707	0.5766	0.5731	0.5679	0.5462	0.5712
L3	0.5771	0.5807	0.5698	0.5716	0.5682	0.5686
L4	0.5636	0.5665	0.5749	0.5694	0.5877	0.5792
L5	0.5725	0.5773	0.5632	0.5673	0.5809	0.5675
avgL1	0.1115	0.1080	0.1156	0.1130	0.1116	0.1100
avgL2	0.1141	0.1153	0.1146	0.1136	0.1092	0.1142
avgL3	0.1154	0.1161	0.1140	0.1143	0.1136	0.1137
avgL4	0.1127	0.1133	0.1150	0.1139	0.1175	0.1158
avgL5	0.1145	0.1155	0.1126	0.1135	0.1162	0.1135
Range value	0.0039	0.0081	0.0030	0.0013	0.0083	0.0049

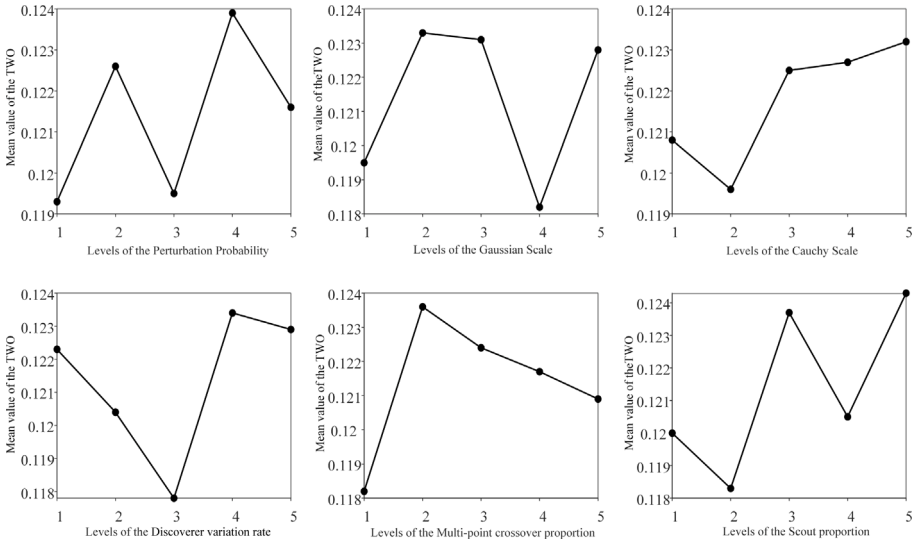


Fig. 6. Mean values of TWO for different level of parameters under small-scale instance.

Table 12
Optimal parameters combination of HSSA.

Instance type	Perturbation probability	Gaussian scale	Cauchy scale	Discoverer variation rate	Multi-point crossover proportion	Scout proportion
Small-scale instance	0.4	0.8	6	0.25	0.05	0.08
Large-scale instance	0.4	0.5	9	0.15	0.08	0.05

of processing sub-lots, where real-time decision-making and solution quality are critical requirements.

To demonstrate the effectiveness of the proposed HSSA in this study, we test the lot-sizing problem with another three algorithms, including differential evolution (DE), improved particle swarm optimization (IPSO) and sparrow search algorithm (SSA), as presented in Table 14. These three comparative algorithms serve dual purposes. On one hand, they are specifically adapted to address the dual-function transportation resource-constrained discrete combinatorial optimization scheduling problem; on the other hand, since our HSSA algorithm incorporates improvements that encompass the strategies and excellent features of these three comparative algorithms, these comparative algorithms are also designed to validate whether utilizing their strengths for enhancement yields effective results. Table 15 shows the statistical

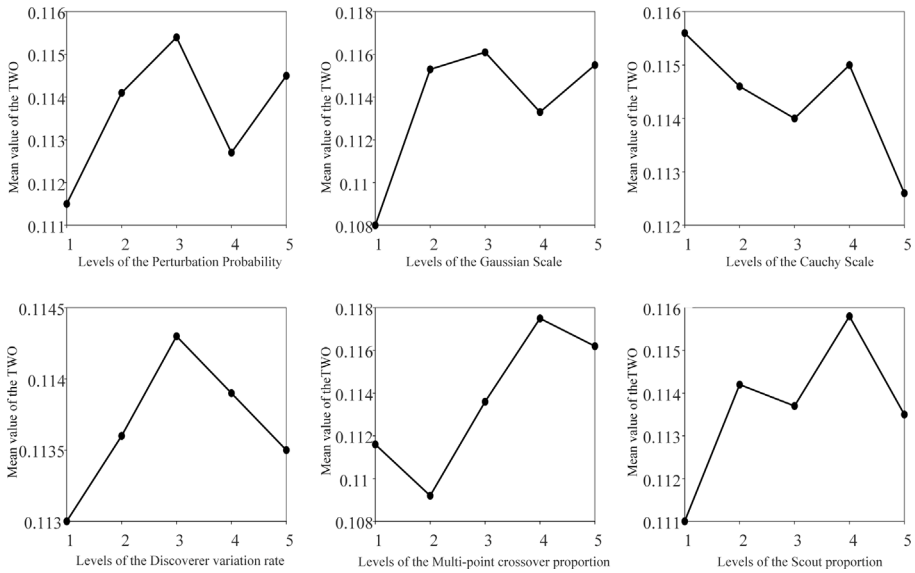


Fig. 7. Mean values of TWO for different level of parameters under large-scale instance.

Table 13
Comparison of the results obtained from Cplex12.10.0.0 and HSSA.

Instance	Number of sub-lots	Cplex12.10.0.0		Optimality gap (%)	HSSA	
		Total weighted objective	t (s)		Total weighted objective	t (s)
data01	10	0.0794	0.66	0	0.0794	0.88
data02	15	0.1609	4.69	0	0.1609	0.86
data03	20	0.1993	21.4	0.45	0.1984	2.34
data04	25	0.2913	7200	34.7	0.1900	2.68
data05	30	0.3157	7200	23.3	0.2420	23.43
data06	35	–	–	–	0.2276	62.53

results of *TWO* values obtained through the four algorithms, including the best objective value obtained for each of the 20 independent runs as well as the minimum and maximum mean values. The minimum and maximum values indicate that the hybrid sparrow search algorithm with multi-policy fusion is more capable of finding the optimal solution as compared with DE, IPSO and SSA.

The four algorithms are compared in terms of iterative trends. Fig. 8 illustrates the iterative processes of the four algorithms for the two different scales. The HSSA exhibits significantly better optimization performance than the other three comparative algorithms. Fig. 9 display the box plots depicting the performance of the four algorithms across two distinct order scales based on twenty independent experimental trials. These experimental results illustrate that the HSSA exhibits a notably lower median value for small-scale instance than the other three benchmark algorithms. The obtained minimum is much smaller than that of the remaining three algorithms, the stability is not inferior to that of the remaining three algorithms. For the large-scale instance, the HSSA is slightly less stable than the remaining three algorithms, however, the best solution found by it is far superior to the remaining three comparative algorithms, and its median is also steadily lower than that of the remaining three comparative algorithms. Consequently, the initialization strategy and multi-strategy framework proposed in this study significantly enhance the solution efficacy of the sparrow search algorithm.

Table 16 compare the optimization effect of each objective of the HSSA approach with multi-strategy fusion, differential evolutionary algorithm, improved PSO algorithm and SSA applied to both small- and large-scale order scenarios. It can be seen that the sparrow search algorithm based on multi-strategy enhancement outperforms the other three comparative algorithms in optimizing the two important indicators of completion time and change mode number, while the total weighted objective is far less than that of the other three algorithms, and it is also comparable to the other three algorithms in terms of product inventory time and extension time. The multi-strategy enhanced HSSA approach demonstrates a marginally longer runtime for both instances than the three baseline algorithms. It achieves notable reductions in completion time and mold change frequency under small-scale order conditions. In comparison, the total order delay and the work-in-process inventory time are not inferior to the other three algorithms. The number of mold changes with the other three algorithms is not inferior to the other three algorithms under large-scale orders. There is a significant reduction in the number of mold changes and completion time for large-scale orders. This is because mold change costs are more substantial based on actual production data, leading to a notable decrease in mold changes. Therefore, the manufacturing cost obtained from optimizing the HSSA approach with multi-strategy fusion is generally better, and the completion time also shows an overall decreasing trend. Moreover, the total weighted objectives of the two order sizes show better results than the remaining three algorithms. Which also proves that the proposed HSSA approach with multi-strategy fusion has better optimization ability for the proposed model.

5. Case study

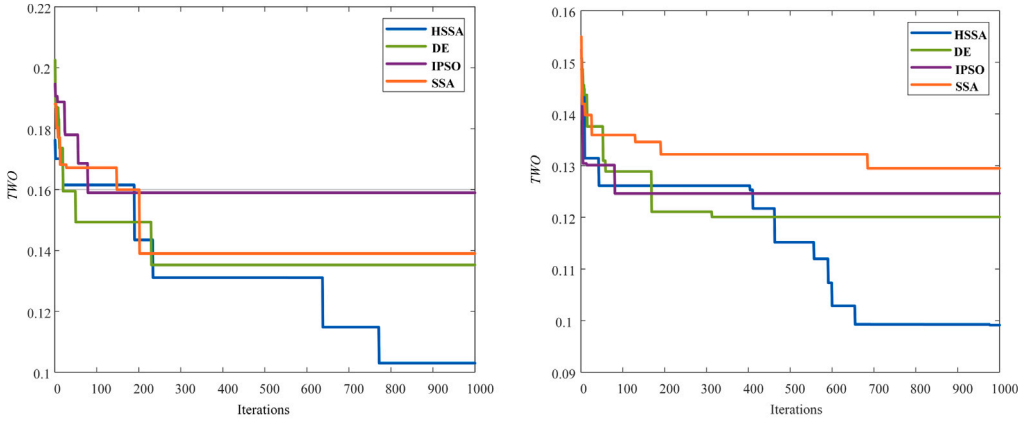
Isolated scheduling approaches cannot meet modern automotive manufacturing demands, where industrial information integration is essential for coordinating production decisions, resource allocation, and scheduling optimization. This research originates from an Advanced Planning and Scheduling Systems (APS) implementation project at a domestic automotive parts manufacturing enterprise. The company primarily engages in the design and development of automotive components including fuel tanks, sunroofs, sheet metal, and resin parts. Field

Table 14
Basic design of the four algorithms.

	HSSA	DE	IPSO	SSA
Method of population initialization	Logistic chaotic mapping	Logistic chaotic mapping	Logistic chaotic mapping	Logistic chaotic mapping
Improvement of iteration operation	Multi-strategy	None	Fusion ant colony algorithm	None

Table 15
Results obtained by 20 independent runs of the four algorithms.

No.	Small scale instance				No.	Large scale instance			
	HSSA	DE	IPSO	SSA		HSSA	DE	IPSO	SSA
1	0.1114	0.1203	0.1503	0.1313	1	0.1133	0.1189	0.1196	0.1182
2	0.1027	0.1447	0.1546	0.1216	2	0.1222	0.1280	0.1232	0.1324
3	0.1395	0.1320	0.1673	0.1488	3	0.1227	0.1249	0.1259	0.1288
4	0.1037	0.1485	0.1682	0.1327	4	0.1224	0.1201	0.1271	0.1293
5	0.1202	0.1313	0.1696	0.1504	5	0.1149	0.1277	0.1223	0.1227
6	0.1030	0.1440	0.1654	0.1496	6	0.1238	0.1204	0.1246	0.1240
7	0.1296	0.1410	0.1549	0.1421	7	0.0991	0.1262	0.1226	0.1215
8	0.1241	0.1487	0.1339	0.1592	8	0.1210	0.1251	0.1183	0.1253
9	0.1235	0.1352	0.1646	0.1501	9	0.1037	0.1227	0.1198	0.1227
10	0.1323	0.1435	0.1682	0.1342	10	0.1192	0.1159	0.1213	0.1269
11	0.1203	0.1360	0.1681	0.1393	11	0.1239	0.1231	0.1252	0.1295
12	0.1205	0.1500	0.1589	0.1390	12	0.1028	0.1204	0.1272	0.1259
13	0.1135	0.1466	0.1549	0.1468	13	0.1236	0.1183	0.1234	0.1314
14	0.1206	0.1395	0.1646	0.1327	14	0.1167	0.1248	0.1234	0.1293
15	0.1046	0.1335	0.1488	0.1309	15	0.1043	0.1204	0.1232	0.1227
16	0.1209	0.1451	0.1668	0.1340	16	0.1180	0.1231	0.1224	0.1258
17	0.1315	0.1401	0.1747	0.1450	17	0.1091	0.1245	0.1192	0.1269
18	0.1297	0.1407	0.1388	0.1327	18	0.1140	0.1186	0.1191	0.1253
19	0.1151	0.1416	0.1661	0.1501	19	0.1046	0.1262	0.1250	0.1288
20	0.1114	0.1422	0.1619	0.1488	20	0.1238	0.1304	0.1227	0.1258
Max	0.1395	0.1500	0.1747	0.1592		0.1239	0.1304	0.1272	0.1324
Min	0.1027	0.1313	0.1339	0.1216		0.0991	0.1159	0.1183	0.1182
Mean	0.1194	0.1417	0.1600	0.1410		0.1152	0.1230	0.1228	0.1262

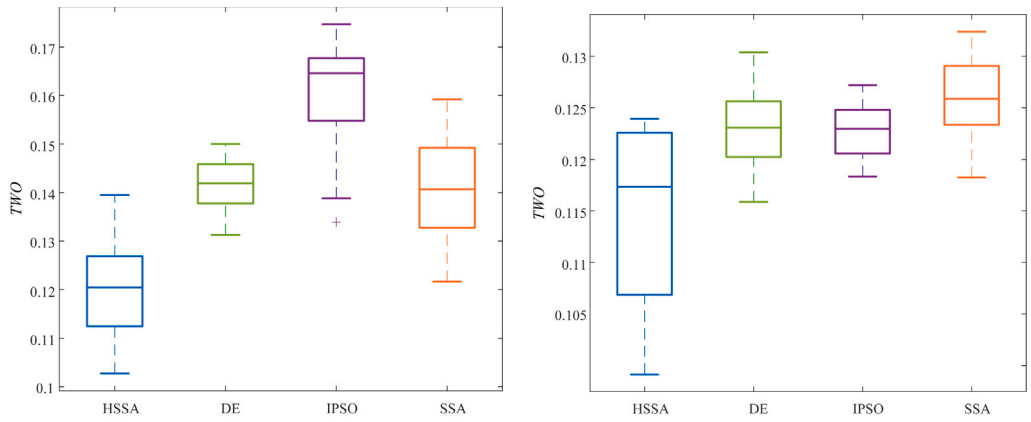


(a) Convergence curves for small-scale instance (b) Convergence curves for large-scale instance

Fig. 8. Convergence curves of *TWO* for the two instances.

Table 16
Performance statistics for different scale of instances.

Instance scale	Algorithm	TT			Makespan			IC			TNS		
		Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean
Small-scale instance	**HSSA**	6.22	0	1.53	137.8	135.9	**136.9**	6.0769E+03	5.9021E+03	6.0034E+03	23	19	**21**
	DE	14.6	0	2.42	139.0	136.3	137.5	6.0549E+03	5.7936E+03	5.9959E+03	24	22	23
	IPSO	25.3	0	7.43	141.3	135.8	137.4	6.2061E+03	5.6737E+03	5.9923E+03	27	22	25
	SSA	8.23	0	0.47	139.0	136.0	137.2	6.1249E+03	5.9110E+03	5.9982E+03	24	21	23
Large-scale instance	**HSSA**	44.8	3.1	23.6	174.6	169.7	**171.7**	8.1032E+03	7.7886E+03	7.9515E+03	48	42	**45**
	DE	55.9	1.8	24.3	176.4	168.0	171.7	8.2687E+03	7.8397E+03	7.9791E+03	50	44	47
	IPSO	18.1	0	4.4	174.7	168.2	172.2	8.1362E+03	7.7289E+03	7.9223E+03	49	45	47
	SSA	54.3	2.7	21.7	174.7	169.4	172.5	8.0650E+03	7.8760E+03	7.9632E+03	49	45	47



(a) Box plots for small-scale instance (b) Box plots for large-scale instance

Fig. 9. Box plots of four algorithms in solving each instance respectively.

investigation revealed that the sunroofs production system constitutes the primary bottleneck constraining overall workshop capacity. According to the enterprise's actual production configuration, the workshop is equipped with three parallel production lines. The sunroofs manufacturing process involves 7 main sequential processes: pre-installing frame assembly, adhesive bonding between glass assembly and frame assembly, installing the motor and wiring harness, sun visor assembly, installing external sealing strip, final assembly and functional testing, with each operation requiring specific processing time and equipment resources.

In the different planning periods, the enterprise may receive small-scale and large-scale customer orders due to the demand fluctuations from host manufacturers, which reflecting the normal changes of the market. Therefore, the workshop needs to make the optimal scheduling in different production scale scenarios. In this paper, we have selected two typical case problems from the actual production containing small-scale problem and large-scale problem. Within the planning horizon, there are 14 different types of sunroofs (i.e. *A*, *B*, *C*, *D*, *E*, *F*, *G*, *H*, *I*, *J*, *K*, *L*, *M*, *N*) need to be produced. After order preprocessing through lot-splitting and lot-merging, sub-lots of sunroofs are released into the production line for processing. The required trolleys for each order can be found in Table 4, with all trolley capacities ranging from 6 to 10 units. The detailed data of the two case problems are given at https://doi.org/10.5281/zenodo.15833291. The proposed HSSA is applied to develop detailed production scheduling solutions for the case problems. The parameters of HSSA are configured for both small-scale and large-scale cases according to Table 12. Table 17 presents the results of the four competitive algorithms. Fig. 10 shows the performance on various objectives obtained by the four algorithms for this two real case problems. The results clearly demonstrate that the HSSA performs excellently in addressing automotive parts parallel production line scheduling problems with dual-function transportation resources. All the experimental result indicate that the proposed mathematical model and HSSA enables scientific decision-making for production scheduling of the auto parts parallel production lines. Moreover, current research provides workshop decision-makers with an effective tool for obtaining optimal schedules among parallel production lines.

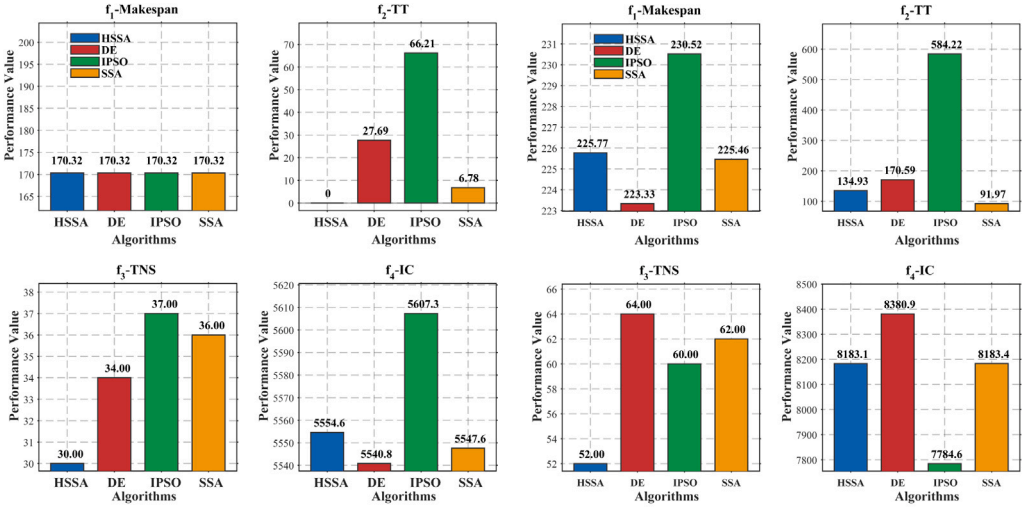
Table 17
Performance of the algorithms for the two real-world case problems.

Problem	Objective	Algorithm			
		HSSA	DE	IPSO	SSA
Small-scale	f_1-Makespan	170.32	170.32	170.32	170.32
	f_2-TT	0	27.69	66.21	6.78
	f_3-TNS	30	34	37	36
	f_4-IC	5.5546E+03	5.5408E+03	5.6073E+03	5.5476E+03
	f-TWO	0.0767	0.1244	0.1639	0.1415
Large-scale	f_1-Makespan	225.77	223.33	230.52	225.46
	f_2-TT	134.93	170.59	584.22	91.97
	f_3-TNS	52	64	60	62
	f_4-IC	8.1831E+03	8.3809E+03	7.7846E+03	8.1834E+03
	f-TWO	0.1563	0.2685	0.2603	0.2488

6. Conclusion

This research discussed the integrated lot-sizing and scheduling problem in an auto parts company, with a particular constraint related to the availability and management of transportation resources during production. A specific multi-objective MIP model is built for the problem with consideration of practical production characteristics, including order delivery constraints, item variety constraints, trolley resource constraints, and mold change constraints. The min–max normalization and linear weight method are employed to transform multi-objective problem into single objectives problem. According to the practical industrial scene, we categorize the test instances into two kinds comprising small-scale instance and large-scale instance. The multi-strategy enhanced HSSA is designed to solve the problem. Taguchi method is used to tune the parameters of the algorithm for different size of problems. Computational experiment and real case study demonstrate substantial superiority of HSSA as compared with the other three candidate algorithms.

With the rapid development of new energy vehicles in recent years, the demand of auto parts like car sunroofs is experiencing explosive growth. This poses great challenges for the decision makers who are engaged in this field. In the future, regarding the complex lot-sizing scheduling problem from auto parts industry, solving it with high quality and efficiency remain crucial, especially in the face of a more dynamic real-time decision-making environment. Thus, as the emerging advanced techniques, such as deep learning, reinforcement learning, transfer learning and knowledge-assisted search methodologies, some attempts at new methods can still be applied to solve the problem, in order to achieve rapid decision-making ability and efficient production. Integrating knowledge-assisted search mechanisms and adaptive learning frameworks could substantially enhance solution quality and system robustness. In addition, more practical conditions can be considered incorporating uncertainties such as demand uncertainty, machine failures and the uncertainty in processing times and setup times, etc. Future endeavors should prioritize algorithm efficiency improvements, model expansion, and the development of adaptive optimization strategies that leverage multiple learning paradigms to effectively address increasingly complex production scenarios.



(a) Performance for small scale problem (b) Performance for large scale problem

Fig. 10. Performance on various objectives obtained by the algorithms for the two real case problems.

CRediT authorship contribution statement

Lei Yue: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Jietao Huang:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation. **Linshan Ding:** Validation, Methodology, Conceptualization. **Xiwen Cai:** Validation, Software, Formal analysis. **Yanming Sun:** Supervision, Project administration, Funding acquisition. **Tao Zou:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data supporting the engineering case study are openly available at https://doi.org/10.5281/zenodo.15833291.

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